An explicit Li-positivity threshold from the numerical verification of the Riemann hypothesis

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October 26, 2026 — draft v4 (v2 after one round of internal refereeing; v3 adds the formalization remark; v4: numerical chain also formalized), not yet externally reviewed

Abstract

Let $\lambda_n = \sum_\rho \bigl[1-(1-1/\rho)^n\bigr]$ be the Li coefficients of the Riemann ξ-function. We prove, without any hypothesis, that $\lambda_n > 0$ for every integer $1 \le n \le N_0 = 1.918\,555\,899\ldots \cdot 10^{24}$ (exact value in Theorem 1). The inputs are the zero-sum formula for λ_n, the Platt–Trudgian verification that all zeros with $|\operatorname{Im}\rho| \le T_0 = 3\cdot 10^{12}$ lie on the critical line, and the explicit zero-counting bound of Hasanalizade, Shen and Wong. The threshold is $N_0 \approx 0.21\,T_0^2$. The statement "verification to height T implies $\lambda_n \ge 0$ for $n \le T^2$" is attributed to an unpublished typescript of Oesterlé (2000); the one published proof with that threshold that we are aware of was retracted, and the published result in this direction, Lagarias (2007), has non-explicit constants and threshold $T^2/(2\log^2 T)$. The present threshold is certified in ball arithmetic, and the argument, including the numerical certificate but not the three inputs, has been checked in Lean 4 with Mathlib.

1 Statement

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, an entire function of order one whose zeros $\rho = \beta + i\gamma$ are exactly the nontrivial zeros of ζ, counted with multiplicity, with $0 < \beta < 1$ and $\gamma \ne 0$ (as $\zeta(\sigma) \ne 0$ for $0 < \sigma < 1$). The multiset of zeros is invariant under $\rho \mapsto \bar\rho$ and $\rho \mapsto 1-\rho$. Li [7] defined

$$\lambda_n = \frac{1}{(n-1)!}\frac{d^n}{ds^n}\Bigl[s^{n-1}\log\xi(s)\Bigr]_{s=1} \qquad (n \ge 1),$$

proved $\lambda_n = \sum_\rho \bigl[1-(1-1/\rho)^n\bigr]$ (sum in the symmetric order $|\gamma| \le T$, $T \to \infty$), and showed that the Riemann hypothesis is equivalent to $\lambda_n \ge 0$ for all $n \ge 1$; see Bombieri and Lagarias [3] for the general form. One has $\lambda_1 = 1 + \gamma_E/2 - \frac{1}{2}\log 4\pi = 0.0230957\ldots$ (here γ_E is Euler's constant).

Theorem 1. *For every integer n with $1 \le n \le N_0 := 1\,918\,555\,899\,886\,294\,190\,260\,224$ one has $\lambda_n > 0$.*

The proof uses exactly the following external inputs.

*Derivation, computation and review were carried out with the assistance of Claude (Anthropic), including an independent adversarial review by a separate model instance. The author takes responsibility for the content.

(I1) The zero-sum formula above. After pairing ρ with $\bar\rho$ the series converges absolutely: $1-(1-1/\rho)^n = n/\rho + \sum_{k=2}^{n}(-1)^{k+1}\binom{n}{k}\rho^{-k}$ and $2\operatorname{Re}(n/\rho) = 2n\beta/|\rho|^2$, so the paired term is $O_n(|\rho|^{-2})$, and $\sum_\rho |\rho|^{-2} < \infty$ (cf. [3]). The symmetric-order sum equals the sum of the paired terms over $\gamma > 0$.

(I2) (Platt–Trudgian [12]) Every zero with $|\gamma| \le T_0 := 3\cdot 10^{12}$ has $\beta = 1/2$. Throughout, T_0 denotes this fixed constant.

(I3) (Hasanalizade–Shen–Wong [4], Cor. 1.2) Let $N(T)$ be the number of zeros with $0 < \gamma \le T$, counted with multiplicity (their proof is by the argument principle applied to ξ). For all $T \ge e$,

$$\big|N(T) - M(T)\big| \le R(T), \qquad M(T) := \frac{T}{2\pi}\log\frac{T}{2\pi e},$$
$$R(T) := 0.1038\log T + 0.2573\log\log T + 9.3675.$$

Since M and R are continuous, the convention at the jumps of N is immaterial; we use the right-continuous N (zeros with $0 < \gamma \le T$), as the Stieltjes integrations below require.

Nothing is assumed about the zeros with $|\gamma| > T_0$ beyond $0 < \beta < 1$. Simplicity of zeros is not used.

2 Notation and elementary lemmas

For a zero ρ write $w_\rho := 1 - 1/\rho = (\rho-1)/\rho$, $r_\rho := |w_\rho|$, $\theta_\rho := \arg w_\rho$. Then

$$r_\rho^2 = \frac{(1-\beta)^2+\gamma^2}{\beta^2+\gamma^2}, \tag{1}$$

so $r_\rho = 1$ if and only if $\beta = 1/2$. Split $\lambda_n = A_n + B_n$, where A_n is the part of the sum (I1) over $|\gamma| \le T_0$ and B_n the part over $|\gamma| > T_0$.

Lemma 2 (on-line pairs)**.** *If $\beta = 1/2$ and $\gamma > 0$, then $\bar\rho = 1-\rho$, $r_\rho = 1$, and the pair $\{\rho,\bar\rho\}$ contributes $2(1-\cos n\theta_\rho) \ge 0$ to λ_n, where $\theta_\rho = 2\arctan\big(1/(2\gamma)\big) \in (0,\pi)$. Consequently*

$$\frac{1}{\gamma} - \frac{1}{12\gamma^3} \le \theta_\rho \le \frac{1}{\gamma}.$$

Proof. $w_\rho = (-\frac12 + i\gamma)/(\frac12 + i\gamma)$ has modulus 1 and equals $(\gamma^2 - \frac14 + i\gamma)/(\gamma^2+\frac14)$, so $\cos\theta = (\gamma^2-\frac14)/(\gamma^2+\frac14)$, $\sin\theta = \gamma/(\gamma^2+\frac14) > 0$ and $\tan(\theta/2) = \sin\theta/(1+\cos\theta) = 1/(2\gamma)$. The conjugate zero gives $\bar w_\rho$, so the pair contributes $2 - 2\operatorname{Re} w_\rho^n = 2(1-\cos n\theta)$. The bounds follow from $x - x^3/3 \le \arctan x \le x$ for $x \ge 0$ with $x = 1/(2\gamma)$. □

Lemma 3 (off-line quadruples)**.** *If $\beta \ne 1/2$, the four distinct zeros $\rho, \bar\rho, 1-\rho, 1-\bar\rho$ (same $|\gamma|$, same multiplicity) contribute together*

$$Q_\rho(n) = 4 - 4\cosh(n\log r_\rho)\cos(n\theta_\rho) \;\ge\; 4 - 4\cosh(n\delta_\rho) = -8\sinh^2(n\delta_\rho/2), \qquad \delta_\rho := |\log r_\rho|.$$

Moreover, if $|\gamma| \ge T_0 \ge 2$, then $\delta_\rho \le c/(2\gamma^2)$ with $c := 1 + T_0^{-2}$.

Proof. $1 - 1/(1-\rho) = -\rho/(1-\rho) = 1/w_\rho$, and similarly for the conjugates. Writing $w_\rho = re^{i\theta}$, the four terms sum to $4 - 2\operatorname{Re}(w^n) - 2\operatorname{Re}(w^{-n}) = 4 - 2(r^n + r^{-n})\cos n\theta = 4 - 4\cosh(n\log r)\cos n\theta$. Since $\cosh \ge 1$ and $\cos n\theta \le 1$ the first inequality follows, and $4 - 4\cosh x = -8\sinh^2(x/2)$. For δ_ρ: by (1), $\log r = \frac12\log(1+x)$ with $x = (1-2\beta)/|\rho|^2$ and $|x| \le 1/\gamma^2 \le 1/4$. For $|x| \le 1/2$, $|\log(1+x) - x| \le x^2/(2(1-|x|)) \le x^2$, so $\delta_\rho \le \frac12\big(|1-2\beta|/|\rho|^2 + \gamma^{-4}\big) \le (1+\gamma^{-2})/(2\gamma^2) \le c/(2\gamma^2)$. □

3 The tail

Lemma 4 (tail lower bound)**.** *For every $n \ge 1$, with $g_n(t) := \sinh^2\big(nc/(4t^2)\big)$,*

$$B_n \;\ge\; -E(n), \qquad E(n) := 4\Big[2R(T_0)g_n(T_0) + \int_{T_0}^{\infty}\Big(\tfrac{1}{2\pi}\log\tfrac{t}{2\pi} + \tfrac{0.1038}{t} + \tfrac{0.2573}{t\log t}\Big)g_n(t)\,dt\Big].$$

In particular, with $u_0 := nc/(4T_0^2)$,

$$E(n) \le E^*(n) := \frac{n^2c^2}{4}\cosh^2(u_0)\,J(T_0),$$

$$J(T_0) := \frac{2R(T_0)}{T_0^4} + \frac{\log(T_0/2\pi)/3 + 1/9}{2\pi T_0^3} + \frac{0.1038}{4T_0^4} + \frac{0.2573}{4T_0^4\log T_0}.$$

Proof. The zeros with $|\gamma| > T_0$ split into on-line conjugate pairs, whose contribution is ≥ 0 by Lemma 2 (whose proof did not use $\gamma \le T_0$), and off-line quadruples. Regrouping the conjugate-paired series into these blocks is legitimate by absolute convergence (I1). By Lemma 3 and monotonicity of $\sinh^2$ on $[0,\infty)$ (as $n\delta_\rho/2 \le nc/(4\gamma^2)$), $B_n \ge -8\sum_{\text{quad}} g_n(\gamma)$. Each quadruple contains exactly two zeros of positive ordinate (ρ and $1-\bar\rho$, both of ordinate γ), so, with multiplicity and since $g_n \ge 0$,

$$\sum_{\text{quad}} g_n(\gamma) \le \tfrac{1}{2}\sum_{\rho:\,\gamma>T_0} g_n(\gamma) = \tfrac{1}{2}\int_{(T_0,\infty)} g_n\,dN.$$

Hence $B_n \ge -4\int_{(T_0,\infty)} g_n\,dN$. Now g_n is C^1 and decreasing with $g_n(t) = O(t^{-4})$, while $N(t) = O(t\log t)$, so Stieltjes integration by parts gives

$$\int_{(T_0,\infty)} g_n\,dN = -g_n(T_0)N(T_0) + \int_{T_0}^{\infty} N(t)\big(-g_n'(t)\big)\,dt.$$

Insert $N(T_0) \ge M(T_0) - R(T_0)$ and, since $-g_n' \ge 0$, $N(t) \le M(t) + R(t) =: N^+(t)$ (both valid by (I3), as $T_0 \ge e$). Integrating by parts once more,

$$\int_{(T_0,\infty)} g_n\,dN \le -g_n(T_0)\big(M(T_0) - R(T_0)\big) + N^+(T_0)g_n(T_0) + \int_{T_0}^{\infty}(N^+)'(t)g_n(t)\,dt,$$

which is the bracket in $E(n)$ because $M'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$ and $R'(t) = 0.1038/t + 0.2573/(t\log t)$. For E^*: $\sinh y \le y\cosh y$ for $y \ge 0$, so $g_n(t) \le (nc/(4t^2))^2\cosh^2 u_0$ for $t \ge T_0$, and $\int_{T_0}^{\infty}\log(t/2\pi)\,t^{-4}dt = (\log(T_0/2\pi)/3 + 1/9)/T_0^3$, $\int_{T_0}^{\infty} t^{-5}dt = 1/(4T_0^4)$, $\int_{T_0}^{\infty} t^{-5}/\log t\,dt \le 1/(4T_0^4\log T_0)$. □

4 The head

Two lower bounds for A_n are used, one for $n \le \pi T_0$ and one for larger n.

Lemma 5 (small-angle head bound)**.** *Let $n \ge 1$ and $\max(n/\pi, 60) \le a \le T_0$. Then*

$$A_n \;\ge\; \frac{4n^2}{\pi^2}\Big(1 - \frac{1}{12a^2}\Big)^2 S^-(a),$$

$$S^-(a) := \frac{M(T_0) - R(T_0)}{T_0^2} - \frac{M(a) + R(a)}{a^2} + \frac{1}{\pi}\Big(\frac{\log(a/2\pi)}{a} - \frac{\log(T_0/2\pi)}{T_0}\Big) - 2\int_a^{T_0} R(t)t^{-3}dt.$$

Proof. By (I2) and Lemma 2, $A_n = \sum_{0<\gamma\le T_0} 2(1-\cos n\theta_\gamma)$ (sum with multiplicity) with every term ≥ 0. Keep only $\gamma > a$. For such γ, $n\theta_\gamma \le n/\gamma < n/a \le \pi$, and $1-\cos x = 2\sin^2(x/2) \ge 2x^2/\pi^2$ for $|x| \le \pi$ (Jordan's inequality). Hence $A_n \ge (4n^2/\pi^2)\sum_{a<\gamma\le T_0}\theta_\gamma^2 \ge (4n^2/\pi^2)(1-1/(12a^2))^2\sum_{a<\gamma\le T_0}\gamma^{-2}$ by Lemma 2. Finally $\sum_{a<\gamma\le T_0}\gamma^{-2} = N(T_0)/T_0^2 - N(a)/a^2 + 2\int_a^{T_0} N(t)t^{-3}dt$; insert $N(T_0) \ge M-R$, $N(a) \le M+R$, $N(t) \ge M-R$ on $[a,T_0]$ (valid as $a \ge 60 > e$), and $2\int M(t)t^{-3}dt = -\frac{1}{\pi}\log(t/2\pi)/t$. □

Lemma 6 (oscillatory head bound)**.** *Let $2\pi \le T_1 < T_0$ and $n \ge 1$. Then*

$$A_n \;\ge\; A'(n,T_1) := 2\big[M(T_0)-M(T_1)-2R(T_0)-2R(T_1)\big] - \frac{4M'(T_0)(T_0^2+\frac14)}{n} - 2nR(T_0)\Big(\frac{1}{T_1}-\frac{1}{T_0}\Big).$$

Proof. With $\theta(t) := 2\arctan(1/(2t))$, decreasing, $\theta'(t) = -1/(t^2+\frac14)$, and all terms nonnegative,

$$A_n \ge \sum_{T_1<\gamma\le T_0} 2(1-\cos n\theta(\gamma)) = 2[N(T_0)-N(T_1)] - 2I, \qquad I := \int_{(T_1,T_0]} \cos(n\theta(t))\,dN(t),$$

the sums being with multiplicity. Riemann–Stieltjes integration by parts (N right-continuous, $\cos n\theta(t)$ of class C^1) gives $I = \cos(n\theta(T_0))N(T_0) - \cos(n\theta(T_1))N(T_1) - \int_{T_1}^{T_0} N(t)\,n\sin(n\theta(t))/(t^2+\frac14)\,dt$. Write $N = M + \tilde S$ on $[T_1,T_0]$ with $|\tilde S| \le R$ by (I3). The M-part reassembles, by ordinary integration by parts, to $I_M = \int_{T_1}^{T_0}\cos(n\theta(t))M'(t)\,dt$; the $\tilde S$-part is bounded by $R(T_0) + R(T_1) + nR(T_0)\int_{T_1}^{T_0} t^{-2}dt$, using R increasing and $1/(t^2+\frac14) \le t^{-2}$. For I_M substitute $t = t(\theta)$, $dt = -(t^2+\frac14)d\theta$: $I_M = \int_{\theta(T_0)}^{\theta(T_1)} \cos(n\theta)\psi(\theta)\,d\theta$ with $\psi(\theta) = M'(t(\theta))(t(\theta)^2+\frac14)$, which is nonnegative and non-increasing in θ because $t(\theta)$ is decreasing and M', t^2 are nonnegative and increasing in t on $[T_1,T_0]$ ($M' \ge 0$ there since $T_1 \ge 2\pi$). By Bonnet's form of the second mean value theorem (nonnegative non-increasing weight, evaluated at the left endpoint), $I_M = \psi(\theta(T_0))\int_{\theta(T_0)}^{\xi}\cos(n\theta)\,d\theta$ for some ξ, so $|I_M| \le 2\psi(\theta(T_0))/n$. Finally $N(T_0)-N(T_1) \ge M(T_0)-M(T_1)-R(T_0)-R(T_1)$. □

Remark 7. *Lemma 5 has no admissible a for $n > \pi T_0$ (then $n/\pi > T_0$); Lemma 6 is what carries the argument to $n \asymp T_0^2$. In Lemma 6 the loss relative to the heuristic T_0^2 threshold comes from estimating $\tilde S = N - M$ by $R(T_0)$ in absolute value; an input of the type $\int_0^T S(t)\,dt = O(\log T)$ (Littlewood), where $S(t) = \pi^{-1}\arg\zeta(\frac12+it)$ differs from $\tilde S$ by a bounded term, would exploit the cancellation instead. The constant* 0.21 *is where $2nR(T_0)/T_1$ crosses $2[M(T_0)-M(T_1)]$ at $T_1 = 2T_0/3$.*

5 Proof of Theorem 1

Let $n_1 \le n \le n_2$ be integers. Put $a := \max(n_2/\pi, 60)$ and, if $a \le T_0$, $A_1(n_1,n_2) := (4n_1^2/\pi^2)(1-1/(12a^2))^2 S^-(a)$; put $A_2(n_1,n_2;T_1) := 2[M(T_0)-M(T_1)-2R(T_0)-2R(T_1)] - 4M'(T_0)(T_0^2+\frac14)/n_1 - 2n_2R(T_0)(1/T_1-1/T_0)$ for $T_1 \in [2\pi, T_0)$.

Lemma 8. *If $\max\{A_1(n_1,n_2), A_2(n_1,n_2;T_1)\} > E^*(n_2)$ (the first term omitted when $a > T_0$), then $\lambda_n > 0$ for every integer $n \in [n_1,n_2]$.*

Proof. $\lambda_n = A_n + B_n \geq A_n - E^*(n)$ by Lemma 4, and E^* is increasing in n (a product of positive increasing factors), so $E^*(n) \leq E^*(n_2)$. If the maximum is attained by A_1: Lemma 5 applies with this a (as $\max(n/\pi, 60) \leq a \leq T_0$ in this case), and since $A_1 > E^*(n_2) > 0$ forces $S^-(a) > 0$, replacing n^2 by n_1^2 only decreases the bound. If attained by A_2: Lemma 6 with $1/n \leq 1/n_1$ and $n \leq n_2$. □

Proof of Theorem 1. Lemma 8 is applied along a chain of 89 consecutive intervals $[n_1, n_2]$ with $n_1 = 1$ initially and each new n_1 equal to the previous n_2, so that every integer in $[1, N_0]$ is covered; T_1 is chosen among $\{T_0/10, T_0/4, T_0/3, T_0/2, 2T_0/3\}$. All quantities are evaluated in ball arithmetic (Arb, via `python-flint` 0.9.0, working precision 60 digits), with $\int_a^{T_0} R(t)t^{-3}\,dt$ replaced by the closed-form upper bound obtained from $\log\log t \leq \log t$, which only decreases $S^-(a)$; for each interval the larger of A_1 and the five candidates $A_2(\cdot; T_1)$ is used. An interval is accepted only when the ball enclosing (head bound) $-E^*(n_2)$ lies entirely in $(0, \infty)$, which is what the Arb comparison with 0 certifies. The last accepted interval is $[1\,918\,555\,885\,591\,938\,922\,971\,136,\ N_0]$, on which the head bound is $4.93658106 \cdot 10^{10}$ against $E^*(N_0) = 4.93658070 \cdot 10^{10}$ (both to within 0.05). The certifying script and its output are archived with this note.[1] □

6 Remarks

Remark 9 (sharpness and the mechanism)**.** *The threshold is $N_0 \approx 0.213\,T_0^2$, attained with $T_1 = 2T_0/3$. The quadruple formula of Lemma 3 shows precisely how an off-line zero affects the Li coefficients: it perturbs λ_n by $-4(\cosh n\delta_\rho - 1)\cos n\theta_\rho$, i.e. by at most about $n^2(1-2\beta)^2/(2|\rho|^4)$ in absolute value until $n \approx \gamma^2/|1-2\beta|$, after which the perturbation grows like $\exp\bigl(n|1-2\beta|/(2\gamma^2)\bigr)$ with oscillating sign. Thus (heuristically, against the main term $\frac{n}{2}\log n$) an off-line zero is detected by the sign of some λ_n only once $n \gtrsim \gamma^2 \log n/|1-2\beta|$; the converse implication is not proved here. In either direction, Li positivity up to N cannot detect a zero close to the line at any height. This is the quantitative content of the detection threshold $n \gtrsim T^2/|t|$ for a zero $\frac{1}{2} + t + iT$ discussed by Voros [13, §1].*

Remark 10 (literature)**.** *Biane, Pitman and Yor [2, p. 441] report, as a private communication of J. Oesterlé, that $\lambda_n \geq 0$ whenever all zeros with $|\operatorname{Im}\rho| < \sqrt{n}$ lie on the line; Voros [13] attributes to the same unpublished typescript the implication with $n \leq T_0^2$. Lagarias [6, Thm. 1.1, eq. (1.17), and the remark preceding it] proves unconditionally $\lambda_n = \frac{n}{2}\log n + c_1 n - \lambda_n(\sqrt{n}) + O(\sqrt{n}\log n)$, where $\lambda_n(T) := \sum_{|\operatorname{Im}\rho|<T}[1-(1-1/\rho)^n]$ is his incomplete Li coefficient (his eq. (1.14); for ζ the dual representation is ζ itself), and deduces the asymptotic $\lambda_n = \frac{n}{2}\log n + c_1 n + O(\sqrt{n}\log n)$ (the form it takes under the Riemann hypothesis) for $n \leq T^2/(2\log^2 T)$ from verification to height T, with non-explicit constants.[2] Omar, Ouni and Mazhouda [9] stated a partial Li criterion with threshold T^2; the paper was retracted [10]. The largest range in which positivity has been certified by direct computation of λ_n is $n \leq 10^5$ (Johansson [5], using Arb). An earlier attempt at an explicit threshold from a zero-free region, Brown [1, Thm. 2], is reported to be defective by Palojärvi [8]. The present note gives, as far as we know, the first explicit and fully proved threshold derived from a verification height, i.e. the first explicit form of the implication attributed to Oesterlé, with constant 0.21 in place of 1; a treatment of $\int \tilde{S}(t)\,dt$ in Lemma 6 exploiting cancellation would likely recover the constant 1.*

[1] `lab/certify_li_threshold.py`, output `log/2026-10-26_certify_li_threshold_v3_output.txt`.

[2] The journal version states $T^2/(2\log^2 T)$; the arXiv version (math/0404394v4) states $T^2/(4\log^2 T)$.

Remark 11 (formalization)**.** *The analytic chain of this note has been checked in Lean 4 with Mathlib (files* `lean/Li*.lean` *in the repository* `https://github.com/s-reiner/rh-notes-2026` *accompanying this note, axioms* `propext`*,* `Classical.choice`*,* `Quot.sound` *only): the pointwise Lemmas 2 and 3 (with the Mathlib-provable constant* $2x^2$ *in the logarithm estimate, hence* $c = 1 + 2T_0^{-2}$*, which gives the same* N_0*), Abel summation over a finite set of ordinates in place of the Stieltjes integrations by parts, the explicit bounds* $E^*(n)$*,* $S^-(a)$ *and* $A'(n, T_1)$ *of Lemmas 4, 5 and 6 (Bonnet's theorem replaced by an integration by parts against* $\sin n\theta$ *with the increasing weight* $M'(t)(t^2 + \frac{1}{4})$*), the reduction of the symmetric zero sum to ordinates* $\gamma > 0$*, and Lemma 8 in the form: for a finite family of zeros closed under* $\rho \mapsto \bar{\rho}$ *and* $\rho \mapsto 1 - \rho$*, with counting function* N *satisfying* $|N - M| \le R$*, the certificate inequality for* $[n_1, n_2]$ *implies* $\sum_\rho \mathrm{Re}[1 - (1 - 1/\rho)^n] > 0$ *for every integer* $n \in [n_1, n_2]$*. The* 89*-interval numerical chain is checked in Lean as well: every certificate inequality is proved from rational enclosures of* $\log 2$*,* $\log 3$*,* $\log 5$ *and* $\log \pi$ *(via Mathlib's series bound for* $\log(1 - x)$ *and its* 20*-digit bounds for* π*), choosing in the head-sum intervals a parameter* $a = 2^i 3^j 5^m \ge \max(n_2/\pi, 60)$ *in place of* n_2/π*, and the binding interval closes with the same margin as the ball-arithmetic run. The resulting Lean theorem states that, for a finite family of zeros as above with all zeros of height* $\le T_0$ *on the line and* $|N - M| \le R$ *on* $[60, T]$*, the partial Li sum is positive for every integer* $1 \le n \le N_0$*. Not formalized: (I1) (the passage from such finite partial sums to* λ_n*), and (I2) and (I3), which enter as hypotheses.*

Remark 12 (the 2026 zero-free half-plane)**.** *The zero-free half-plane* $\mathrm{Re}\, s > 7/8$ *announced in [11] (unrefereed; an accompanying Lean formalization is reported, which we have not checked) would allow* $c = 3/4 + T_0^{-2}$ *in Lemma 3; it changes* N_0 *by* 0.25% *because the threshold is set by the head bound, and we have not used it.*

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